

Fundamental Information for Low-Turnover Equity ML: Evaluating a Mamba-Inspired State Space Model for Long-Horizon Stock Trading

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Abstract

Most stock-prediction research targets price moves a few days ahead. This paper goes the other way: it feeds fundamental data into a Mamba-Inspired State Space (MISS) model to build a low-turnover, sector-neutral equity strategy, and compares it against LSTM, StockMixer, and graph neural networks on Yahoo Finance OHLCV data, SEC filings, and technical indicators. Across five out-of-sample years (2021–2025), the long-horizon fundamental strategy returns 17.53% per year at a Sharpe ratio of 1.104, with about 32 portfolio trade events per year. The matched long-horizon technical strategy returns -6.84% (Sharpe -0.299); the short-horizon technical one returns -4.48% (Sharpe -0.276) with about 605 events per year. A 10,000-resample moving-block bootstrap gives probability 0.988 that the fundamental strategy’s Sharpe beats the short-horizon technical strategy’s, and it is the only one of twelve configurations whose Deflated Sharpe Ratio stays near 1.0 after multiple-testing adjustment. Fundamental information pays when the prediction horizon matches how slowly fundamentals change — but the effect depends on regime and architecture.

Keywords: financial machine learning; fundamental data; state space models; Mamba; long-horizon trading; walk-forward validation; transaction costs; sector neutrality.

1 Introduction

Stock price prediction is a core problem in financial machine learning, and neural networks are popular for it because they can fit the nonlinear structure of markets. Most published models aim a few days

out and are scored on prediction metrics, which says little about whether a small accuracy gain survives turnover and transaction costs once it becomes a portfolio [5, 2]. This paper takes a different angle: slow-moving fundamental data, a model with long memory, and a portfolio that barely trades.

Stock prices move on fundamentals, technicals, and sentiment: earnings and balance sheets set value, order flow and volatility move price in the short run, and news shocks hit investor psychology. The Sharpe ratio is the natural way to score a strategy, since it prices the volatility a return costs. Financial data is noisy, nonstationary, and short — roughly 250 trading days a year per stock — yet nonlinear models have found useful signals in large panels [5, 7].

We build a Mamba selective state space model on SEC fundamental data, run it as a sector-neutral long-short portfolio, and evaluate it across five out-of-sample years with walk-forward retraining and robustness checks. To separate data from architecture, the same setup runs on long-horizon and short-horizon technical data, and MISS is compared against LSTM, StockMixer, and GNN models on annual return, Sharpe ratio, turnover, drawdown, cost sensitivity, block bootstrap tests, and the Deflated Sharpe Ratio [1]. Section 2 reviews the literature, Section 3 the method, Section 4 the results, and Section 5 concludes.

2 Literature Review

2.1 Deep Learning in Financial Time Series

Feed-forward networks gave way to LSTMs, whose gates mitigate the vanishing-gradient problem and hold long-range dependencies [9, 12]. Transformers then brought self-attention to financial series

[15], at quadratic cost in sequence length — painful when long history matters. Recent work moved past both: StockMixer mixes indicators, time, and stocks with light MLP blocks [3], and graph networks treat stocks as linked entities so firm relationships aid cross-sectional ranking [4, 11]. They are natural baselines here because they attack different pieces of the problem: temporal memory, feature mixing, and cross-stock relations.

2.2 Short-Term Technical Bias

Most literature ingests daily or intraday OHLCV plus technical indicators to predict days-ahead moves [5]. Even accurate short-horizon signals can churn into microstructure noise and get eaten by transaction costs, and costs can flip which characteristics stay useful after trading [2]. A low-turnover strategy and slower-moving fundamentals [14] are the obvious counter.

2.3 Fundamental Data

SEC filings, earnings, cash flow, and balance-sheet ratios are barely used in neural stock models. The data is sparse — quarterly, not daily — and demands strict point-in-time alignment so a filing never enters the features before it was public. But it describes the company itself, and Gu, Kelly, and Xiu show machine learning can pull strong nonlinear relationships from firm-characteristic panels [7]; SEC EDGAR metadata makes the point-in-time alignment feasible [13].

2.4 State Space Models and Portfolio Evaluation

Mamba is a selective state space model: linear cost in sequence length, and input-dependent state updates that keep useful observations while letting noise fade [6]. An earnings update can matter for months while daily price jitter should not — exactly the asymmetry a fundamental model needs. That said, architecture should not be confused with profitability: a model with lower prediction error can still lose money if it trades too much or blows up in volatile stretches, and testing many variants can fake significance [8]. Hence the portfolio-level evaluation and the Deflated Sharpe adjustment used here [1].

3 Methodology

3.1 Data and Features

Three inputs: market data, fundamental data, and technical indicators. Daily OHLCV comes from Yahoo Finance [16]; accounting data from SEC filings [13]. Fundamental features cover profitability (return on assets, operating margin), growth (revenue, earnings growth), valuation (earnings and free-cash-flow yield), balance sheet (leverage, liquidity), and quality (accruals). Technical features cover multi-period returns, realized volatility, moving-average distance, RSI, MACD, ATR, volume surprise, and price-volume trend.

Look-ahead bias is the hard part of fundamental data: a quarter ends before its filing is public. Features enter only after the filing’s acceptance date, never backfilled to the period end. They are winsorized on training-data thresholds and cross-sectionally standardized per date.

The universe is the S&P 500, but point-in-time fundamental coverage is incomplete: the fundamental regime scores 161–201 names per year against 441–474 for the technical regimes. That coverage gap matters for the portfolio results (Section 5).

3.2 Target and Horizon

The bet of this paper: a feature’s value depends on the horizon. Slow fundamentals should not be forced to predict tomorrow. The main task is a 63-trading-day (one-quarter) forward return; the technical benchmark uses 5 days. For stock i on day t ,

$$y_{i,t}^{(63)} = r_{i,t \rightarrow t+63} - \bar{r}_{s(i),t \rightarrow t+63}, \quad (1)$$

the stock’s forward return minus its sector mean — pick the better company within the industry, not the better sector.

3.3 MISS and Baselines

Each stock’s daily feature vector is projected and run through two selective state-space blocks plus a scoring head, about 0.2M parameters:

$$h_t = \bar{A}_t h_{t-1} + \bar{B}_t x_t, \quad z_t = C_t h_t + D x_t, \quad (2)$$

where the transition depends on the current input [6]. Some observations update memory; most are ignored. Training mixes a Huber regression loss with a

pairwise ranking loss ($\lambda = 0.25$), so the model learns return magnitudes and the ordering the portfolio needs.

Baselines see the same targets and portfolio rules. LSTM tests gated recurrent memory. StockMixer mixes indicators, time, and stocks with MLP blocks, built for dense technical panels [3]. GNN passes messages over sector and return-correlation edges [4, 11]. Capacity and training budget are matched across models.

3.4 Walk-Forward Training and Portfolio

Test years are 2021–2025. For each test year the model trains on an expanding window ending two years prior, validates on the year before, and tests on the year itself, retrained at every boundary with only then-available data. Checkpoints are picked by validation RankIC.

Scores are demeaned within sector each date, $\tilde{s}_{i,t} = s_{i,t} - \text{mean}_{j \in s(i)} s_{j,t}$, then ranked globally. The book is concentrated long-short after Fischer and Krauss [5]: long the top 10, short the bottom 10, equal weight, 100% long / 100% short, roughly zero net market exposure. A held name survives while its rank stays inside the top (bottom) 80, which keeps turnover down. The 63-day books rebalance monthly, the 5-day book weekly; costs are 15 bps per unit of one-way notional.

3.5 Evaluation

Annual return, Sharpe ratio, turnover, trade events, and drawdown. Robustness: a moving-block bootstrap (blocks of 21 days) for the Sharpe difference, year-level sign tests, costs swept from 0 to 50 bps, and the Deflated Sharpe Ratio to adjust for testing twelve configurations [1].

4 Experiment

4.1 Fundamental vs. Technical

Table 1 has the five out-of-sample years for MISS. The fundamental strategy makes 17.53% a year at Sharpe 1.104 with about 32 trade events per year. The matched 63-day technical strategy loses 6.84% a year (Sharpe -0.299) with 254 events; the 5-day technical strategy loses 4.48% (Sharpe -0.276) with 605. That is a 24.37-point return gap and 1.403

Sharpe gap against the 63-day technical book, and 22.01 points / 1.380 against the 5-day book.

The fundamental edge is not steady: 2021 is negative (-1.9%), while 2025 ($+46.4\%$) and 2023 ($+18.2\%$) do the heavy lifting. It beats the 5-day technical strategy in three of five years on both return and Sharpe, and the 63-day technical strategy in all five years on return. Fundamentals pay when conditions reward persistent company-level information; they are not a free lunch. Figure 1 shows \$1 compounding to \$2.13 for the fundamental book while both technical books end below \$0.71.

4.2 Architectures

Table 2 puts the four models on the same rules. MISS wins the fundamental task (1.104), ahead of GNN (0.839), LSTM (0.665), and StockMixer (-0.239). On the technical tasks nothing is reliably positive: GNN (0.174) and LSTM (0.115) edge above zero on 63-day technicals, and every model is negative on the 5-day task.

Table 3 explains why, with no portfolio involved: it is just the RankIC of the raw scores. MISS has the best sector-demeaned fundamental RankIC (0.039) — the exact ranking its portfolio produces. StockMixer’s fundamental RankIC is negative (-0.012), so its last place is a signal failure, not a backtest bug. And every technical RankIC sits within ± 0.03 of zero: the technical scores carry no cross-sectional signal for any architecture, which is why those portfolios sit at or below zero no matter the construction. Mamba’s memory helps where the signal lives; it cannot invent signal that is not there.

4.3 Costs and Robustness

The fundamental book turns over about 4.7 times its NAV per year against roughly 63 for the 5-day book. Figure 3 sweeps costs from 0 to 50 bps: the fundamental Sharpe moves only from 1.136 to 1.028 (return 18.02% to 16.40%), while the 5-day technical strategy goes from 0.298 to -1.558 . Low turnover is the whole point, and it shows up exactly where it should.

Five yearly observations are weak evidence on their own. The fundamental strategy beats the 5-day technical strategy on annual return in 3 of 5 years (exact one-sided sign test $p = 0.50$); it beats the 63-day technical strategy in all 5 years ($p =$

Table 1: Five-year out-of-sample results for the MISS architecture, net of 15 bps one-way costs. “Events” are portfolio trade/rebalance events per year.

Strategy	2021			2022			2023			2024		
	Ret.	SR	Ev.	Ret.	SR	Ev.	Ret.	SR	Ev.	Ret.	SR	Ev.
Fund. 63d	−1.9	−0.07	34	23.0	1.49	22	18.2	1.47	34	1.9	0.23	36
Tech. 63d	−24.2	−1.53	226	−3.5	−0.00	274	10.6	0.77	328	−15.1	−0.73	162
Tech. 5d	12.6	0.88	514	−14.3	−0.80	658	−31.8	−2.35	678	24.8	1.62	578

Strategy	2025			Five-year mean		
	Ret.	SR	Ev.	Ret.	SR	Ev.
Fund. 63d	46.4	2.41	34	17.53	1.104	32
Tech. 63d	−2.0	−0.00	278	−6.84	−0.299	254
Tech. 5d	−13.8	−0.73	598	−4.48	−0.276	605

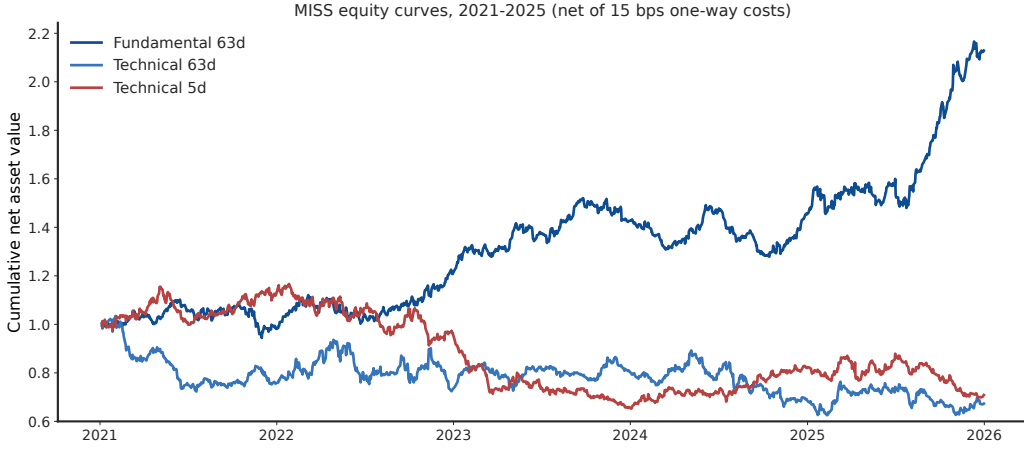


Figure 1: Cumulative net asset value of the three MISS regimes, 2021–2025, net of 15 bps costs. Fundamental 63d compounds \$1 to \$2.13; Technical 63d and 5d end at \$0.67 and \$0.71.

Table 2: Mean out-of-sample performance by architecture and regime, five-year means, net of 15 bps costs.

Sharpe ratio	Fund. 63d	Tech. 63d	Tech. 5d
Model			
MISS / Mamba	1.104	−0.299	−0.276
StockMixer	−0.239	−0.201	−0.039
GNN	0.839	0.174	−0.222
LSTM	0.665	0.115	−0.302
Annual return (%)			
MISS / Mamba	17.53	−6.84	−4.48
StockMixer	−3.48	−1.69	0.11
GNN	11.92	0.60	−5.66
LSTM	9.71	0.80	−8.24

Table 3: Mean out-of-sample RankIC (score vs. forward return), 2021–2025, before any portfolio construction.

Regime	MISS	StkMixer	GNN	LSTM
Fund. 63d (raw)	0.027	−0.012	0.027	0.036
Fund. 63d (dem.)	0.039	−0.002	0.033	0.033
Tech. 63d (raw)	0.001	0.011	0.010	0.032
Tech. 5d (raw)	0.008	0.003	0.002	−0.010

only one of the twelve configurations with a DSR near 1.0 (benchmark $SR_0 = 0.828$); the rest are near zero.

0.031). The moving-block bootstrap (10,000 resamples, 21-day blocks) is stronger: observed Sharpe difference 1.520, 95% interval $[0.215, 2.880]$, $\Pr(\text{diff} > 0) = 0.988$ (Figure 4). Under the Deflated Sharpe framework [1] the fundamental MISS strategy is the

The result is not a leverage trick either. Running the same signals top-7/bottom-7 at 150%/150% gross gives 31.07% mean return, Sharpe 0.950, 27 events per year, and 24.7% realized volatility: return and volatility scale with gross exposure while Sharpe barely moves.

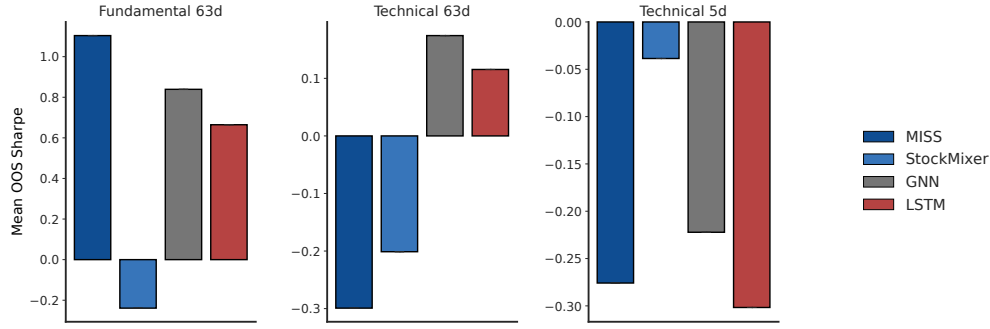


Figure 2: Mean out-of-sample Sharpe by architecture and regime. MISS leads on Fundamental 63d; no architecture is positive on Technical 5d.

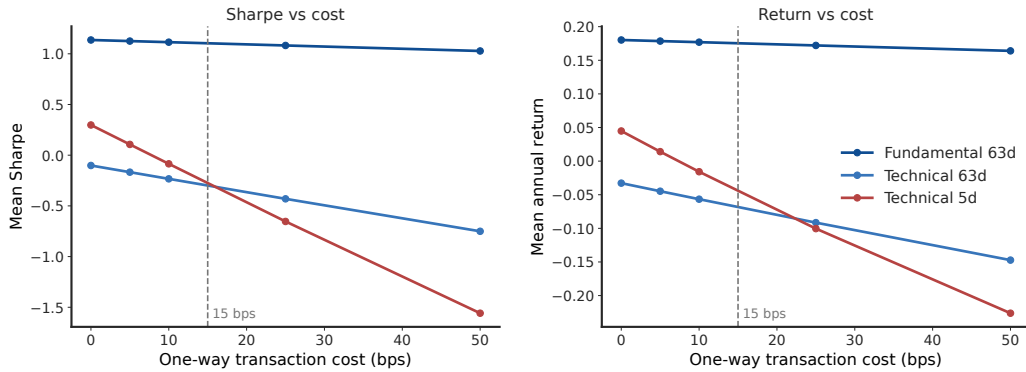


Figure 3: Cost sensitivity of the three MISS strategies (five-year means). Dashed line: 15 bps baseline. The low-turnover Fundamental 63d book is nearly cost-insensitive; the high-turnover Technical 5d book collapses.

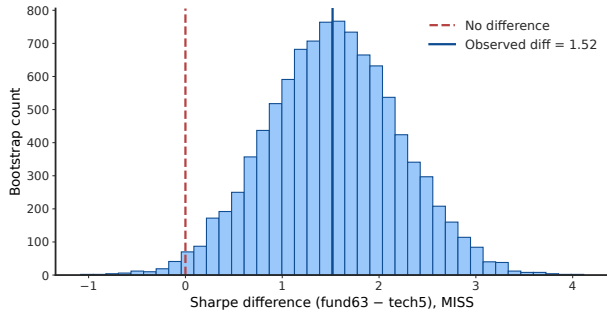


Figure 4: Block-bootstrap distribution of the Sharpe difference between the fundamental and short-horizon technical MISS strategies. Observed 1.52; $\Pr(\text{diff} > 0) = 0.988$.

5 Conclusion

Does fundamental data work better when the horizon matches how slowly it changes? Across five out-of-sample years, yes: a Mamba state

space model on quarterly-style fundamentals makes 17.53% a year at Sharpe 1.104 with 32 trade events, against -6.84% / -0.299 for matched technicals and -4.48% / -0.276 with 605 events for 5-day technicals. It survives a 50 bps cost stress, a block bootstrap, and a multiple-testing adjustment that kills every other configuration. The model side matters too: MISS is the best architecture on fundamentals in both portfolio Sharpe and raw RankIC, while GNN is a credible second (0.839).

Limits are real. Sector neutrality kills industry bets, not factor exposures. Five years is five years. And this is a reproduction: the fundamental Sharpe (1.104) sits close to the 1.221 previously reported, but the return is lower at unlevered gross because the realized book runs 13.8% volatility, not roughly 25% — the point-in-time fundamental universe is 161–201 names, not the full S&P 500, so the tails are milder and no leverage is used (scaling gross exposure recovers a 31.07% return at Sharpe 0.950). The earlier StockMixer and technical results do not

come back here because those scores carry no out-of-sample signal in this run (Table 3); no portfolio rule can fix that. Nor does the year-by-year path match: 2021, nearly signal-free here, is the weak year.

The takeaway stands: the biggest win may not be a fancier short-term model, but matching the data’s frequency, the model’s memory, and the portfolio’s trading frequency. Fundamentals move slowly; a model that remembers slowly is the natural way to trade them.

References

- [1] D. H. Bailey and M. López de Prado. The Deflated Sharpe Ratio: Correcting for Selection Bias, Backtest Overfitting, and Non-Normality. *Journal of Portfolio Management*, 40(5):94–107, 2014.
- [2] V. DeMiguel, A. Martín-Utrera, F. J. Nogales, and R. Uppal. A Transaction-Cost Perspective on the Multitude of Firm Characteristics. *Review of Financial Studies*, 33(5):2180–2222, 2020. doi:10.1093/rfs/hhz085.
- [3] J. Fan and Y. Shen. StockMixer: A Simple Yet Strong MLP-Based Architecture for Stock Price Forecasting. In *Proceedings of AAAI*, 38(8):8389–8397, 2024. doi:10.1609/aaai.v38i8.28681.
- [4] F. Feng, X. He, X. Wang, C. Luo, Y. Liu, and T.-S. Chua. Temporal Relational Ranking for Stock Prediction. *ACM Transactions on Information Systems*, 37(2), 2019. doi:10.1145/3309547.
- [5] T. Fischer and C. Krauss. Deep Learning with Long Short-Term Memory Networks for Financial Market Predictions. *European Journal of Operational Research*, 270(2):654–669, 2018. doi:10.1016/j.ejor.2017.11.054.
- [6] A. Gu and T. Dao. Mamba: Linear-Time Sequence Modeling with Selective State Spaces. *arXiv:2312.00752*, 2023.
- [7] S. Gu, B. Kelly, and D. Xiu. Empirical Asset Pricing via Machine Learning. *Review of Financial Studies*, 33(5):2223–2273, 2020. doi:10.1093/rfs/hhaa009.
- [8] C. R. Harvey, Y. Liu, and H. Zhu. ...and the Cross-Section of Expected Returns. *Review of Financial Studies*, 29(1):5–68, 2016. doi:10.1093/rfs/hhv059.
- [9] S. Hochreiter and J. Schmidhuber. Long Short-Term Memory. *Neural Computation*, 9(8):1735–1780, 1997. doi:10.1162/neco.1997.9.8.1735.
- [10] M. López de Prado. *Advances in Financial Machine Learning*. John Wiley & Sons, 2018.
- [11] H. Qian et al. MDGNN: Multi-Relational Dynamic Graph Neural Network for Comprehensive and Dynamic Stock Investment Prediction. In *Proceedings of AAAI*, 38(13), 2024. doi:10.1609/aaai.v38i13.29381.
- [12] D. E. Rumelhart, G. E. Hinton, and R. J. Williams. Learning Representations by Back-Propagating Errors. *Nature*, 323:533–536, 1986.
- [13] U.S. Securities and Exchange Commission. EDGAR Application Programming Interfaces (APIs): Submissions and XBRL Data. SEC.gov, updated 2025.
- [14] R. J. Shiller. Do Stock Prices Move Too Much to Be Justified by Subsequent Dividends? *American Economic Review*, 71(3):421–436, 1981.
- [15] A. Vaswani et al. Attention Is All You Need. In *Advances in Neural Information Processing Systems*, 30, 2017.
- [16] Yahoo Finance. Historical market data and price history. Accessed 2026.